Hands-on Experience with Docker Data Persistence - Exercise

In this train, we'll gain hands-on experience with real-world scenarios to reinforce learning and application of data persistence within Docker.

Questions

Question 1

What is the recommended method for persisting data in Docker?

1. Using Docker Volumes
2. Storing data inside the container filesystem
3. Using temporary bind mounts
4. Keeping data in environment variables

View answer and explanation

- Correct answer: Using Docker Volumes
- Docker volumes are the preferred method for data persistence because they exist outside the container filesystem, ensuring data is not lost when a container stops or is removed.

Question 2

Which command would you use to create a named volume for persistent storage?

1. docker volume create my_data
2. docker run --volume new_data:/app
3. docker persist volume my_data
4. docker storage create my_data

View answer and explanation ✖

- Correct answer: docker volume create my_data
- This command explicitly creates a named volume that can be reused across multiple containers.

Question 3

How can you mount a volume when running a container?

1. docker run -v my_volume:/app/data my_image
2. docker start --persist my_volume my_image
3. docker run --save-volume my_volume my_image
4. docker mount volume=my_volume my_image

View answer and explanation ✖

- Correct answer: docker run -v my_volume:/app/data my_image
- The `-v` flag allows you to specify a volume when starting a container, ensuring the data persists outside the container lifecycle.

Question 4

What happens to a bind mount if the container is deleted?

1. The data remains intact on the host
2. The data is deleted along with the container
3. The data becomes inaccessible
4. Docker moves the data to a backup folder

View answer and explanation ✖

- Correct answer: The data remains intact on the host

- Bind mounts use the host's filesystem directly, meaning the data persists even if the container is removed.

Question 5

Which of the following is NOT an advantage of using Docker Volumes?

1. They persist data beyond the container lifecycle
2. They can be shared between multiple containers
3. They automatically back up data to an external server
4. They allow separation of storage from the container

View answer and explanation ✖

- Correct answer: They automatically back up data to an external server
- Docker volumes provide persistent storage but do not automatically back up data to external servers.

Question 6

Your team is developing a web application using Docker containers. They need to ensure that uploaded user files persist even if the application container is restarted or replaced. What is the best way to achieve this?

1. Store the files inside the container's /uploads directory
2. Use a temporary volume
3. Mount a Docker volume at /uploads
4. Rely on environment variables to track file locations

View answer and explanation ✖

- Correct answer: Mount a Docker volume at /uploads
- Docker volumes persist beyond container lifecycles, making them the best solution for ensuring user files remain available after container restarts.

Question 7

A database container is deployed, but upon restart, all the stored data is lost. What could be the reason?

1. The container is using ephemeral storage instead of a volume
2. The database is misconfigured
3. The image does not support persistent storage
4. The container is running in interactive mode

View answer and explanation ✖

- Correct answer: The container is using ephemeral storage instead of a volume
- By default, container storage is ephemeral. Data will be lost unless a volume is mounted at the database's data directory.

Question 8

A DevOps engineer needs to share logs between multiple containers running on the same host. Which of the following approaches would be most effective?

1. Write logs to a file inside each container
2. Use a named Docker volume to share log files
3. Copy log files manually between containers
4. Restart the containers to sync logs

View answer and explanation ✖

- Correct answer: Use a named Docker volume to share log files
- A named volume allows multiple containers to write and read log files from a shared location, ensuring persistence and accessibility.

Question 9

An e-commerce application runs inside a container and needs to retain shopping cart data for users. What would be the best approach?

1. Use a volume to store cart data
2. Store the cart data inside container memory
3. Rely on the container's built-in storage
4. Save cart data inside a temporary directory

View answer and explanation ✖

- Correct answer: Use a volume to store cart data
- Volumes ensure that user data, such as shopping cart information, persists across restarts, preventing loss of user progress.

Question 10

A startup deploys a containerised application that processes large amounts of customer data. They need to back up this data daily to prevent loss. What is the best way to manage this?

1. Manually copy files from the container
2. Schedule backups from a mounted volume
3. Store all data inside the container filesystem
4. Use environment variables to track changes

View answer and explanation ✖

- Correct answer: Schedule backups from a mounted volume
- Backing up from a volume ensures that data is regularly saved and can be restored in case of failures.